

Exercise J01

Q1. There are n A.M's between 3 and 54, such that the 8th mean: $(n - 2)^{\text{th}}$ mean:: 3: 5. The value of n is.

- (A) 12 (B) 16 (C) 18 (D) 20

Ans. B

Sol. $\frac{(54 - 3)}{n + 1} = d$; $d = \frac{51}{n + 1}$

$$\frac{A_8}{A_{n-2}} = \frac{3}{5} \Rightarrow \frac{3 + 8 \frac{51}{n+1}}{3 + (n-2) \frac{51}{n+1}} = \frac{3}{5} \Rightarrow \frac{3n + 3 + 408}{3n + 3 + 51n - 102} = \frac{3}{5}$$

$$\Rightarrow 15n + 2055 = 162n - 297 \Rightarrow 147n = 2352 ; n = 16$$

Q2. Sum of the series

$$S = 1^2 - 2^2 + 3^2 - 4^2 + \dots - 2002^2 + 2003^2 \text{ is}$$

- (A) 2007006 (B) 1005004 (C) 2000506 (D) none of these

Ans. A

Sol. $S = 1^2 - 2^2 + 3^2 - 4^2 - 4^2 + \dots - 2002^2 + 2003^2 = \sum_{n=1}^{2003} n^2 - 2 [2^2 + 4^2 + \dots + 2002^2]$

$$= \sum_{n=1}^{2003} n^2 - 2 \times 4 = \frac{2003 \times 2004 \times 4007}{6} - \frac{8(1001)(1002)(2003)}{6}$$

$$= \frac{2003 \times 2008}{6} [4007 - 4004] = 2007006 \text{ Ans.}$$

Q3. If $1, \log_9 (3^{1-x} + 2), \log_3 (4 \cdot 3^x - 1)$ are in A.P. then x equals

- (A) $\log_3 4$ (B) $1 - \log_3 4$ (C) $1 - \log_4 3$ (D) $\log_4 3$

Ans. B

Sol. -

Q4. If the third term of an A.P. is 7 and its 7th term is 2 more than three times of its 3rd term, then sum of its first 20 terms is-

- (A) 228 (B) 74 (C) 740 (D) 1090

Ans. C

Sol. Let a be the first term and d be the common difference of the given A.P. It is given that

$$a_3 = 7 \text{ and } a_7 = 3a_3 + 2$$

$$\Rightarrow a + 2d = 7 \text{ and } a + 6d = 3(a + 2d) + 2$$

$$\Rightarrow a + 2d = 7 \text{ and } a = -1$$

$$\Rightarrow a = -1, d = 4$$

$$\therefore S_{20} = \frac{20}{2} \{2 \times -1 + (20 - 1) \times 4\}$$

$$\left[U \sin g : S_n = \frac{n}{2} \{2a + (n - 1)d\} \right]$$

$$\Rightarrow S_{20} = \frac{20}{2} (-2 + 76) = 740$$

Q5. Let T_r be the r^{th} term of an A.P. whose first term is a and common difference is d . If for some positive integers $m, n, m \neq n, T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then $a - d$ equals-

(A) 0 (B) 1 (C) $1/mn$ (D) $\frac{1}{m} + \frac{1}{n}$

Ans. A

Sol. -

Q6. Let $a_1, a_2, a_3, \dots$ be terms of an A.P. If $\frac{a_1 + a_2 + \dots + a_p}{a_1 + a_2 + \dots + a_q} = \frac{p^2}{q^2} = , p \neq q$ then $\frac{a_6}{a_{21}}$ equals -

(A) $\frac{7}{2}$ (B) $\frac{2}{7}$ (C) $\frac{11}{41}$ (D) $\frac{41}{11}$

Ans. C

Sol. $\therefore \frac{(S_m)_{IAP}}{(S_n)_{IIAP}} = \frac{m^2}{n^2}$ then $\frac{T_m}{T_n} = \frac{2m - 1}{2n - 1}$

so $\frac{a_6}{a_{21}} = \frac{2 \times 6 - 1}{2 \times 21 - 1} = \frac{11}{41}$

Q7. The number of natural numbers < 300 that are divisible by 6 but not by 9 is :

(A) 49 (B) 37 (C) 33 (D) 16

Ans. C

Sol. -

Q8. A person is to count 4500 currency notes. Let a_n denote the number of notes he counts in the n^{th} minute. If $a_1 = a_2 = \dots a_{10} = 150$ and $a_{11}, \dots$ are in AP with common difference of -2 , then the time taken by him to count all notes is :-

(A) 24 minutes (B) 34 minutes (C) 125 minutes (D) 135 minutes

Ans. B

Sol. $4500 = 150 \cdot 10 + \{ 148 + 146 + \dots \text{upto } n \text{ terms} \}$

$$= 1500 + \frac{n}{2} \{296 + (n-1)(-2)\}$$

$$\Rightarrow n^2 - 149n + 3000 = 0$$

$$\Rightarrow (n-24)(n-125) = 0$$

$$\Rightarrow n = 24 \quad \because n \neq 125$$

So total time taken = $10 + 24 = 34$ min.

Q9. Let S_1 be the sum of first $2n$ terms of an arithmetic progression. Let S_2 be the sum of first $4n$ terms of the same arithmetic progression. If $(S_2 - S_1)$ is 1000, then the sum of the first $6n$ terms of the arithmetic progression is equal to :

(A) 7000 (B) 5000 (C) 3000 (D) 1000

Ans. C

$$\text{Sol. } S_1 = \frac{2n}{2}(2a + (2n-1)d) = n(2a + (2n-1)d)$$

$$S_2 = \frac{4n}{2}(2a + (4n-1)d) = 2n(2a + (4n-1)d)$$

$$S_2 - S_1 = 1000$$

$$2n(2a + (4n-1)d) - n(2a + (2n-1)d) = 1000$$

$$n[4a + (4n-1)2d - 2a - (2n-1)d] = 1000$$

$$n[2a + d(8n-2-2n+1)] = 1000$$

$$n[2a + d(6n-1)] = 1000$$

$$\text{Sum of } 6n \text{ terms} = \frac{6n}{2}(2a + (6n-1)d) = 3000$$

$$S = 3000$$

Q10. Let a_n be the n^{th} term of an A.P. If $\sum_{r=1}^{100} a_{2r} = \alpha$ and $\sum_{r=1}^{100} a_{2r-1} = \beta$, then the common difference of the A.P. is :-

(A) $\frac{\alpha - \beta}{200}$ (B) $\alpha - \beta$ (C) $\frac{\alpha - \beta}{100}$ (D) $\beta - \alpha$

Ans. C

$$\text{Sol. } \sum_{r=1}^{100} a_{2r} = a_2 + a_4 + a_6 + \dots + a_{200} = \alpha$$

$$= (a + d) + (a + 3d) + \dots + (a + 199d) = \alpha$$

$$\sum_{r=1}^{100} a_{2r-1} = \beta = a_1 + a_3 + \dots + a_{199} = \beta$$

$$= a + (a + 2d) + \dots + (a + 198d) = \beta$$

$$\frac{100}{2} [a + d + a + 199d] = \beta$$

$$\Rightarrow 50 (2a + 200d) = \alpha \quad \dots\dots\dots(1)$$

$$\frac{100}{2} [a + a + 198d] = \beta$$

$$\Rightarrow 50 (2a + 198d) = \beta \quad \dots\dots\dots(2)$$

$$\alpha - \beta = 50 (2d)$$

$$= d = \frac{\alpha - \beta}{100}$$

Q11. For any three positive real numbers a, b and c, $9 (25a^2 + b^2) + 25 (c^2 - 3ac) = 15b (3a + c)$. Then

- (A) a, b and c are in A.P. (B) a, b and c are in G.P. (C) b, c and a are in G.P.
(D) b, c and a are in A.P.

Ans. D

Sol. $x^2 + y^2 + z^2 = xy + yz + zx$

$$\Rightarrow x = y = z$$

$$15a = 3b = 5c$$

$$a = \frac{b}{5} = \frac{c}{3} = \lambda$$

$$a = \lambda$$

$$b = 5\lambda$$

$$c = 3\lambda$$

Q12. Let $a_1, a_2, a_3, \dots, a_{49}$ be in A.P. such that $\sum_{k=0}^{12} a_{4k+1} = 416$ and $a_9 + a_{43} = 66$. If

$a_1^2 + a_2^2 + \dots + a_{17}^2 = 140m$ then m is equal to

- (A) 34 (B) 33 (C) 66 (D) 68

Ans. A

Sol. Let $a_1 = a$ common difference = d

$$\sum_{k=0}^{12} a_{4k+1} = 416 \quad a + 24d = 32$$

$$a_9 + a_{43} = 66 \Rightarrow a + 25d = 33$$

$$a = 8$$

$$a_1^2 + a_2^2 + \dots + a_{17}^2 = 8^2 + 9^2 + \dots + 24^2$$

$$= (1^2 + 2^2 + \dots + 24^2) - (1^2 + 2^2 + \dots + 7^2)$$

$$4760 = 140 \text{ m}$$

Q13. If $x > 1$, $y > 1$, $z > 1$ are in GP, then $\frac{1}{1 + \ln x}$, $\frac{1}{1 + \ln y}$, $\frac{1}{1 + \ln z}$ are in:

(A) AP (B) HP (C) GP (D) None of the above

Ans. B

Sol. -

Q14. For a sequence $\{a_n\}$, $a_1 = 2$ and $\frac{a_{n+1}}{a_n} = \frac{1}{3}$. Then $\sum_{r=1}^{20} a_r$ is

(A) $\frac{20}{2}[4 + 19 \times 3]$ (B) $3\left(1 - \frac{1}{3^{20}}\right)$ (C) $2(1 - 3^{20})$ (D) None of these

Ans. B

Sol. $a_1 = 2$; $a_{n+1} = \frac{a_n}{3}$; $a_2 = \frac{a_1}{3} = \frac{2}{3}$; $a_3 = \frac{a_2}{3} = \frac{2}{3^2}$

$$a_1 + a_2 + \dots + a_{20} = 2 + \frac{2}{3} + \frac{2}{3^2} + \dots = \frac{2 \cdot \left[1 - \left(\frac{1}{3}\right)\right]}{1 - \frac{1}{3}} = 3 \left(1 - \frac{1}{3^{20}}\right)$$

Q15. α, β be the roots of the equation $x^2 - 3x + a = 0$ and γ, δ the roots of $x^2 - 12x + b = 0$ and numbers

α, β, δ, d (in this order) form an increasing G.P., then

(A) $a = 3, b = 12$ (B) $a = 12, b = 3$ (C) $a = 2, b = 32$ (D) $a = 4, b = 16$

Ans. C

Sol. $\alpha + \beta = 3$, $\alpha\beta = a$; $\gamma + \delta = 12$, $\gamma\delta = b$

α, β are in G.P. let r be the common ratio so $(1 + r) = 3$

$$\alpha r^2 (1 + r) = 12 \Rightarrow r^2 = 4 \Rightarrow r = 2$$

$$\text{so } \alpha = 1 \Rightarrow \text{so } a = 2, b = 32 \quad \text{Ans.}$$

Q16. If the sum of n terms of a G.P. (with common ratio r) beginning with the p^{th} term is k times the sum of an equal number of terms of the same series beginning with the q^{th} term, then the value of k is:

(A) $r^{p/q}$ (B) $r^{q/p}$ (C) r^{p-q} (D) r^{p+q}

Ans. C

Sol. $\frac{ar^{p-1}(r^n - 1)}{r - 1} = k \cdot \frac{ar^{q-1}(r^n - 1)}{r - 1}$; $r^{p-1} = k \cdot r^{q-1}$; $k = r^{p-q}$

Q17. In an increasing geometric series, the sum of the second and the sixth term is $\frac{25}{2}$ and the product of the third and fifth term is 25. Then, the sum of 4th, 6th and 8th terms is equal to :

- (A) 35 (B) 30 (C) 32 (D) 26

Ans. A

Sol. Let first term = a and common ratio = r.

$$\text{Given } ar + ar^5 = \frac{25}{2} \quad \dots(1)$$

$$\text{and } ar^2 \cdot ar^4 = 25 \Rightarrow ar^3 = 5 \quad \dots(2)$$

$$\text{From (1) \& (2) ; } r = \sqrt{2} \text{ and } a = \frac{5}{2\sqrt{2}}.$$

$$\begin{aligned} t_4 + t_6 + t_8 &= ar^3 + ar^5 + ar^7 \\ &= ar^3 (1 + r^2 + r^4) \\ &= 5 (1 + 2 + 4) = 35 \end{aligned}$$

Q18. In a geometric progression consisting of positive terms, each term equals the sum of the next two terms. Then the common ratio of this progression equals-

- (A) $\frac{1}{2}(1 - \sqrt{5})$ (B) $\frac{1}{2}\sqrt{5}$ (C) $\sqrt{5}$ (D) $\frac{1}{2}(\sqrt{5} - 1)$

Ans. D

$$\text{Sol. } a = ar + ar^2 \quad \Rightarrow \quad r^2 + r - 1 = 0 \quad \Rightarrow r = \frac{-1 \pm \sqrt{5}}{2}$$

$$\Rightarrow r = \frac{\sqrt{5} - 1}{2} \text{ (-ve not permissible)}$$

Q19. Three positive numbers form an increasing G.P. If the middle term in this G.P. is doubled, the new numbers are in A.P. Then the common ratio of the G.P. is

- (A) $2 - \sqrt{3}$ (B) $2 + \sqrt{3}$ (C) $\sqrt{2} + \sqrt{3}$ (D) $3 + \sqrt{2}$

Ans. B

$$\text{Sol. } a \quad a_1 \quad ar^2 \quad \rightarrow \text{ G.P. } ; \quad a \quad 2ar \quad ar^2 \quad \rightarrow \text{ A.P.}$$

$$L_1 \quad ar = a + ar^2 \quad ; \quad 4r = r + r^2$$

$$r^2 - 4r + 1 = 0 \quad ; \quad r = \frac{4 \pm 2\sqrt{3}}{2} = 2 + \sqrt{3}, 2 - \sqrt{3}$$

$$\text{But } r > 1 \quad \therefore r = 2 + \sqrt{3}$$

Q20. If m is the A.M. of two distinct real numbers l and n ($l, n > 1$) and G_1, G_2 and G_3 are three geometric means between l and n, then $G_1^4 + 2G_2^4 + G_3^4$ equals.

- (A) $4 l^2 mn$ (B) $4 l m^2 n$ (C) $4 l mn^2$ (D) $4 l^2 m^2 n^2$

Ans. B

Sol. $m = \frac{l+n}{2}$

$$G_1 = l \left(\frac{n}{l} \right)^{\frac{1}{4}}$$

$$G_2 = l \left(\frac{n}{l} \right)^{\frac{2}{4}}$$

$$G_3 = l \left(\frac{n}{l} \right)^{\frac{3}{4}}$$

$$G_1^4 + 2G_2^4 + G_3^4$$

$$= l^3 n + 2l^2 n^2 + l n^3$$

$$= l n (l^2 + 2ln + n^2)$$

$$= l n (l + n)^2$$

$$= 4 l m^2 n$$

Q21. If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is :

(A) 1 (B) 7/4 (C) 8/5 (D) 4/3

Ans. D

Sol. $T_2 = a$
 $T_5 = a + 3d$
 $T_9 = a + 7d$
 $(a + 3d)^2 = a(a + 7d)$

$$d = \frac{a}{9}$$

$$r = \frac{a + 3d}{a} = 1 + \frac{3d}{a}$$

$$= \frac{4}{3}$$

Q22. If A, G and H are respectively the A.M., G.M. & H.M. of three positive numbers a, b, and c, then the equation whose roots are a, b, and c is given by :

(A) $x^3 - 3Ax^2 + 3G^3x - G^3 = 0$ (B) $x^3 - 3Ax^2 + 3(G^3/H)x - G^3 = 0$

(C) $x^3 + 3Ax^2 + 3(G^3/H)x - G^3 = 0$ (D) $x^3 - 3Ax^2 - 3(G^3/H)x + G^3 = 0$

Ans. B

Sol. $AM = A = \frac{a + b + c}{3}$; $GM = G = (abc)^{1/3}$

$$HM = H = \frac{3abc}{ab + bc + ca} = \frac{3G^3}{ab + bc + ca}$$

Equation whose roots are a,b,c $\Rightarrow x^3 - (a + b + c)x^2 + (\Sigma ab)x - abc = 0$

$$\Rightarrow x^3 - 3Ax^2 + \frac{3G^3}{H}x - G^3 = 0 \text{ Ans}$$

- Q23. If $x_i > 0$, $i = 1, 2, \dots, 50$ and $x_1 + x_2 + \dots + x_{50} = 50$, then the minimum value of $\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}}$ equal to
 (A) 50 (B) $(50)^2$ (C) $(50)^3$ (D) $(50)^4$

Ans. A

Sol. $x_1 + x_2 + x_3 + \dots + x_{50} = 50$

AM $\geq$ HM

$$\frac{x_1 + x_2 + \dots + x_{50}}{50} \geq \frac{1}{\frac{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_{50}}}{50}}$$

$$\Rightarrow \frac{x_1 + x_2 + \dots + x_{50}}{50} \geq \frac{50}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}}} \Rightarrow \frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}}$$

- Q24. If A_1, A_2 be two A.M.s and G_1, G_2 be two G.M.s between a and b , then $\frac{A_1 + A_2}{G_1 G_2}$ is equal to

(A) $\frac{a+b}{2ab}$ (B) $\frac{2ab}{a+b}$ (C) $\frac{a+b}{ab}$ (D) $\frac{a+b}{\sqrt{ab}}$

Ans. C

Sol. -

- Q25. If P, Q be the A.M., G.M. respectively between any two rational numbers a and b , then $P - Q$ is equal to

(A) $\frac{a-b}{a}$ (B) $\frac{a+b}{2}$ (C) $\frac{2ab}{a+b}$ (D) $\left(\frac{\sqrt{a} - \sqrt{b}}{\sqrt{2}}\right)^2$

Ans. D

Sol. $P = \frac{a+b}{2}, Q = \sqrt{ab}; P - Q = \frac{a+b - \sqrt{ab}}{2} = \frac{(\sqrt{a} - \sqrt{b})^2}{2}$

- Q26. If $\alpha \in \left(0, \frac{\pi}{2}\right)$ then $\sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}$ is always greater than or equal to:

(A) $2 \tan \alpha$ (B) 1 (C) 2 (D) $\sec^2 \alpha$

Ans. A

Sol. -

- Q27. The harmonic mean of the roots of the equation $(5 + \sqrt{2})x^2 - (4 + \sqrt{5})x + 8 + 2\sqrt{5} = 0$ is

- (A) 2 (B) 4 (C) 6 (D) 8

Ans. B

Sol. $\alpha + \beta = \frac{4 + \sqrt{5}}{5 + \sqrt{2}}$
 $\alpha \beta = \frac{8 + 2\sqrt{5}}{5 + \sqrt{2}}$
 $\text{H.M.} = \frac{2\alpha\beta}{\alpha + \beta} = \frac{2(8 + 2\sqrt{5})}{(4 + \sqrt{5})} = 4.$

Q28. If $a_1, a_2, \dots, a_n$ are in H.P., then the expression $a_1a_2 + a_2a_3 + \dots + a_{n-1}a_n$ is equal to –

- (A) $(n-1)(a_1 - a_n)$ (B) na_1a_n (C) $(n-1)a_1a_n$ (D) $n(a_1 - a_n)$

Ans. C

Sol. $\frac{1}{a_1}, \frac{1}{a_2}, \dots, \frac{1}{a_n}$ are in A.P.
 $\Rightarrow \frac{1}{a_2} - \frac{1}{a_1} = \dots = \frac{1}{a_n} - \frac{1}{a_{n-1}} = d$
 $\Rightarrow a_1a_2 = \frac{a_1 - a_2}{d}, a_2a_3 = \frac{a_2 - a_3}{d}, \dots, \frac{a_{n-1} - a_n}{d} = a_{n-1} - a_n$
 $\Rightarrow a_1a_2 + a_2a_3 + \dots + a_{n-1}a_n = \frac{a_1 - a_2}{d} + \dots + \frac{a_{n-1} - a_n}{d} = \frac{a_1 - a_n}{d} = (n-1)a_1a_n$

Q29. If $x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$ where a, b, c are in A.P. and $|a| < 1, |b| < 1, |c| < 1$ then

- (A) GP (B) AP (C) Arithmetic - Geometric Progression (D) HP

Ans. D

Sol. $x = \frac{1}{1-a}, y = \frac{1}{1-b}, z = \frac{1}{1-c} \Rightarrow a, b, c$ are in A.P.
 $\Rightarrow 1-a, 1-b, 1-c$ are also in A.P.
 $\Rightarrow \frac{1}{1-a}, \frac{1}{1-b}, \frac{1}{1-c}$ are in H.P.

Q30. The sum of infinite series $1 - \frac{3}{2} + \frac{5}{4} - \frac{7}{8} + \dots$ is-

- (A) $2/9$ (B) $2/3$ (C) $-2/9$ (D) $9/2$

Ans. A

Sol. -

Q31. If $(10)^9 + 2(11)^1 (10)^8 + 3(11)^2 (10)^7 + \dots + 10(11)^9 = k(10)^9$, then k is equal to:

- (A) 100 (B) 110 (C) 121/10 (D) 441/100

Ans. A

Sol. Let $s = (10)^9 + 2(11)^1 (10)^8 + 3(11)^2 (10)^7 + \dots + 10(11)^9$

$$\therefore \frac{11}{10} s = (11)^1 (10)^8 + 2(11)^2 (10)^7 + \dots + 9(11)^9 + (11)^{10}$$

subtract

$$\left(1 - \frac{11}{10}\right)s = (10)^9 + (11)^1 (10)^8 + (11)^2 (10)^7 + \dots + (11)^9 - (11)^{10}$$

$$\Rightarrow -\frac{1}{10} s = \frac{10^9 \left\{1 - \left(\frac{11}{10}\right)^{10}\right\}}{1 - \frac{11}{10}} - (11)^{10} \Rightarrow -\frac{1}{10} s = 10^9 \frac{\left\{10^{10} - 11^{10}\right\}}{10^{10}} \times \frac{1}{-}$$

$$\Rightarrow -\frac{1}{10} s = -10^{10} + 11^{10} - 11^{10} \therefore s = 10^{11}$$

$$\therefore \text{given } 10^{11} = k(10)^9 \therefore k = 100$$

Q32. The sum of the series $1^3 - 2^3 + 3^3 - \dots + 9^3 =$

- (A) 300 (B) 125 (C) 425 (D) 0

Ans. C

Sol. -

Q33. If $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$ upto $\infty = \frac{\pi^2}{6}$, then $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots =$

- (A) $\pi^2/12$ (B) $\pi^2/24$ (C) $\pi^2/8$ (D) None of these

Ans. C

Sol. $I = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \infty = \frac{\pi^2}{6}$

Let $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = A$

$$I = \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \infty\right) + \frac{1}{2^2} \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \infty\right)$$

$$\Rightarrow I = A + \frac{I}{4} \Rightarrow A = \frac{3I}{4} = \frac{3}{4} \times \frac{\pi^2}{6} \Rightarrow A = \frac{\pi^2}{8}$$

Q34. The sum of first 9 terms of the series $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \dots$ is:

- (A) 71 (B) 96 (C) 142 (D) 192

Ans. B

Sol.
$$T_n = \frac{1^3 + 2^3 + \dots + n^3}{1 + 3 + 5 + \dots + (2n-1)} = \frac{\frac{n(n+1)^2}{4}}{n^2}$$
$$= \frac{(n+1)^2}{4}$$
$$\sum T_n = \frac{1}{4}(2^2 + 3^2 + \dots + 10^2)$$
$$= 96$$

Q35. If the sum of the first ten terms of the series $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \dots$, is $\frac{16}{5}m$, then m is equal to :

(A) 100 (B) 99 (C) 102 (D) 101

Ans. D

Sol.
$$\frac{16m}{5} = \sum_{r=2}^{11} \left(\frac{4r}{5}\right)^2$$
$$\frac{16m}{5} = \frac{16}{25} \sum_{r=2}^{11} r^2 = \frac{16}{25} \times 505$$
$$m = 101$$

Q36. If, for a positive integer n, the quadratic equation, $x(x+1) + (x+1)(x+2) + \dots + (x+n-1)(x+n) = 10n$ has two consecutive integral solutions, then n is equal to

(A) 10 (B) 11 (C) 12 (D) 9

Ans. B

Sol.
$$\sum_{r=1}^n (x+r-1) = 10n$$
$$nx^2 + n^2x + \sum r(r-1) = 10n$$
$$nx^2 + n^2x + \frac{n(n^2-1)}{3} = 10n$$
$$x^2 + nx + \frac{n^2-31}{3} = 0$$
$$|\alpha - \beta| = \left| \frac{\sqrt{D}}{a} \right| = 1$$
$$\Rightarrow n = 11$$

Q37. Let A be the sum of the first 20 terms and B be sum of the first 40 terms of the series $1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \dots$. If $B - 2A = 100\lambda$, then λ is equal to

(A) 464 (B) 496 (C) 232 (D) 248

Ans. D

Sol. $f(x) = 1^2 + 2(2)^2 + 3^2 + \dots + 2(x)^2$
 $(1^2 + 2^2 + \dots + n^2) + 2(2^2 + 4^2 + \dots + n^2)$

$$\frac{n(n+1)(2n+1)}{6} + 4 \frac{\left(\frac{1}{2}\left(\frac{n}{2}+1\right)(n+1)\right)}{6}$$

$$f(x) = \frac{n(n+1)^2}{2};$$

$$B - 2A = f(40) - 2f(20)$$

$$= \frac{40 + (41)^2}{2} - 2 \left(\frac{20 \times (21)^2}{2} \right);$$

$$100 \lambda = 24800$$

$$\lambda = 248$$

Q38. The coefficient of x^8 in the polynomial $(x-1)(x-2)(x-3) \dots (x-10)$ is :

(A) 2640 (B) 1320 (C) 1370 (D) 2740

Ans. B

Sol. $f(x) = (x-1)(x-2)(x-3) \dots (x-9)(x-10)$

$$x^{10} - [1 + 2 + \dots + 10]x^9 + (1 \cdot 2 + 1 \cdot 3 + 1 \cdot 4 + \dots)x^8 + \dots$$

$$\text{Coefficient of } x^8 = (1 \cdot 2 + 1 \cdot 3 + \dots + 1 \cdot 10) + (2 \cdot 3 + \dots + 2 \cdot 10) + (3 \cdot 4 + \dots + 3 \cdot 10) + \dots + (9 \cdot 10)$$

$$(1 + 2 + 3 + \dots + 10)^2 = (1^2 + 2^2 + \dots + 10^2) + 2[(1 \cdot 2 + \dots + 1 \cdot 10) + \dots + (9 \cdot 10)]$$

$$\left(\frac{10 \times 11}{2}\right)^2 = \frac{10 \times 11 \times 21}{6} + 2[\text{Coefficient of } x^8]$$

$$\frac{2640}{2} = \text{coefficient of } x^8$$

$$\text{Coeff. of } x^8 = 1320$$

Q39. Find the value of $\frac{2}{1^3} + \frac{6}{1^3 + 2^3} + \frac{12}{1^3 + 2^3 + 3^3} + \frac{20}{1^3 + 2^3 + 3^3 + 4^3} + \dots$
 upto ∞ terms :

(A) 2 (B) $\frac{1}{2}$ (C) 4 (D) $\frac{1}{4}$

Ans. C

Sol. $S = \frac{2}{1^3} + \frac{6}{1^3 + 2^3} + \frac{12}{1^3 + 2^3 + 3^3} + \frac{20}{1^3 + 2^3 + 3^3 + 4^3} + \dots$

$$T_n = \frac{n(n+1)}{\left(\frac{n(n+1)}{2}\right)^2} = \frac{4}{n(n+1)}$$

$$S = \sum_{n=1}^{\infty} \frac{4}{n(n+1)} = 4 \sum_{n=1}^{\infty} \frac{(n+1) - n}{n(n+1)}$$

$$S = 4 \sum_{n=1}^{\infty} \left[\frac{1}{n} - \frac{1}{n+1} \right] = 4$$